THE VIEW AT END OF Q3 2026

The Enterprise GenAI Stack

Veyan®

The new baseline set by this review: reference architecture and product landscape for a regulated UK/EU asset manager · evidence as of 9 October 2026

9 layers, L1 → L9 | **8** enterprise controls | **138** products assessed | **1,449** sources | **58** Strategic · **67** Tactical · **13** Experimental

Strategic — platform default (most carry a condition) | Tactical — a stated niche or estate | Experimental — pilot only, outside regulated paths | AWS Strategic only where that cloud is primary

CONTROL PLANE — firm-owned policy and one evidence store, around every call

C1 AI traffic gateway — model, tool (MCP) and agent (A2A) calls

- **LiteLLM** — Enterprise licence · pin and sign builds
- **Kong AI Gateway** — where Kong is the API standard
- Google Cloud — **Apigee AI gateway** — MCP support GA
- Azure — **Azure APIM AI gateway** — GA policies only
- AWS — **AgentCore Gateway** — MCP gateway
- **Portkey** — Palo Alto Networks · Prisma AIRS
- **agentgateway** — MCP / A2A on Kubernetes
- **Envoy AI Gateway** — Agent Router · in-cluster
- **Cloudflare AI Gateway** — non-confidential work, BYOK

C2 Guardrails — policy and tests owned by the firm

- AWS — **Bedrock Guardrails** — standalone API from the gateway
- Google Cloud — **Model Armor** — strict residency
- Azure — **Azure AI Content Safety** — Prompt Shields on documents
- **NeMo Guardrails** — orchestrator · still 0.x
- **Llama Guard / Prompt Guard** — one detector among several
- **Guardrails AI** — Harvey-owned · plan migration

C3 Privacy service (DLP / PII) — one API, six enforcement points

- **Presidio** — behind a firm privacy-service API
- Google Cloud — **Sensitive Data Protection** — Google's DLP service
- Azure — **Purview DSPM for AI** — posture, not prompt-path DLP
- **Skyflow** — vault tokenisation
- **Protegrity** — existing customers

C4 Identity and authorisation — every agent a registered identity

- **OPA** — policy decision point · CNCF
- **SPIFFE / SPIRE** — workload identity · CNCF
- **Cedar** — policy language · AgentCore Policy
- **Okta / Auth0 for AI Agents** — where Okta holds the workforce
- **Entra Agent ID** — where Entra holds the workforce
- **MCP Authorization** — Anthropic-originated · alt: OAuth on OpenAPI

C5 Configuration of record — prompts, pins and manifests in Git

- **LangSmith Prompts** — with LangSmith as L9 platform
- **Langfuse Prompts** — with Langfuse as L9 platform
- **Prompts as code (Git)** — the configuration of record
- **PromptLayer** — framework-neutral registry
- **LaunchDarkly AgentControl** — rollout between approved variants

C6 AI FinOps — cost per task, budgets fail closed

- **FinOps FOCUS** — open billing data standard
- **Gateway cost attribution** — budgets that fail closed
- **Vantage** — reporting layer only
- **CloudZero** — reporting layer only
- **Helicone** — maintenance mode · migrate

C7 AI security — secrets, supply chain, sandboxing

- **HashiCorp Vault** — IBM · agentic IAM · BUSL
- **Model & package scanning** — safetensors by default
- **OpenSSF Model Signing** — sign weights you produce
- **Prisma AIRS** — Palo Alto estates
- **HiddenLayer** — independent specialist
- **Check Point AI Guardrails** — runtime detector · self-host for client data

C8 Model risk and governance — inventory, validation, evidence store

- **OpenLineage** — + firm GenAI facets
- **IBM watsonx.governance** — IBM estates
- **Credo AI** — policy-led programmes
- **Collibra AI Governance** — where Collibra is the catalogue
- **ModelOp** — after full due diligence
- **ValidMind** — model-risk-led firms

MODEL PLANE

L1 Foundation-model portfolio

two unrelated vendors + small + open-weight, all pinned

A two-vendor mid tier qualified on one suite, a small tier and a self-hosted open-weight tier; every version pinned.

- **OpenAI GPT-6** — Astra · Sol · Luna · 6.1 Sol
- **Mistral** — Medium 3.5 · Large 3 · EU-hosted
- **Anthropic Claude** — hyperscaler EU route · tier set by reader
- **Gemma 4** — small open-weight tier · Apache 2.0
- Google Cloud — **Google Gemini 3.x** — pin versions · short lifetimes
- **Grok 4.7 (SpaceXAI)** — via a hyperscaler only
- **DeepSeek V4** — weights or in-tenant only
- **Z.ai GLM-5.3** — MIT Flash weights · after sanctions review
- **Meta Muse / Llama** — block the contributor tier
- **Qwen 3.8** — self-host by policy
- **Kimi K3** — custom licence

L2 Inference & model access

in-region access; vLLM as the private exit route

The primary cloud's model service in an approved UK/EU region; one open-weight model on vLLM as the tested exit route.

- PATTERN · NOT SCORED — **Primary cloud's model service** — Bedrock · Foundry · Gemini Enterprise Agent Platform, in region
- **vLLM** — default engine · the exit route
- **Fireworks AI** — once ISO certificates confirmed
- **SGLang** — backup engine once CVE is fixed
- **Hugging Face Hub** — governed open-weight supply
- **Together AI** — EU dedicated, ZDR on
- **OpenRouter** — behind the gateway · Stripe deal
- **Ollama** — developer tier
- **Cerebras** — low latency, non-confidential
- **LM Studio** — desktops only · no service use
- **llm-d** — CNCF sandbox · pilot
- **NVIDIA Dynamo** — beta · pilot

AGENT PLANE

L3 Orchestration: workflows & agents

deterministic by default, durable, approval gates

Workflows by default with one bounded model step, durable execution, and approval interrupts only a named human can resume.

- **LangGraph** — default · workflows + agents
- **Temporal** — durable execution
- **Microsoft Agent Framework** — GA · Microsoft and .NET estates
- **Pydantic AI** — typed agent steps
- Google Cloud — **Google ADK** — on Agent Engine
- AWS — **Strands + AgentCore Runtime** — framework + managed runtime
- **OpenAI Agents SDK** — sandboxed sub-step · pre-1.0
- **CrewAI** — use Flows for regulated work
- **LlamaIndex** — retrieval toolkit
- **Vercel AI SDK** — TypeScript app tier
- **Claude Agent SDK** — Alpha · sandboxed only · alt: LangGraph
- **Mistral Agents API** — Workflows in beta

L4 Tools & connectivity

nothing reachable except through the governed gateway

Read-only tools for authoritative data behind a tool-governance sub-layer: private registry, pinned definitions, policy, audit.

- **A2A 1.0** — cross-team delegation · signed cards
- **MCP** — behind a governed gateway · alt: OpenAPI
- AWS — **AgentCore Gateway + Identity** — managed tool governance
- **Browserbase** — only where no API exists
- **E2B** — microVM sandbox
- **Agent Skills** — script-free, internal · alt: AGENTS.md
- **Exa** — via egress proxy + DLP
- **Tavily** — Nebius-owned
- **Composio** — May 2026 token incident · self-host

KNOWLEDGE PLANE

L5 Memory service (part of L6)

policy-gated writes, erasure by person

A governed record class stored in L6: write gate, subject index, erasure and a snapshot per run. Built last.

- AWS — **AgentCore Memory** — inside its own runtime
- Google Cloud — **Memory Bank** — Agent Engine
- **Zep / Graphiti** — Graphiti self-hosted
- **Mem0** — OSS behind a firm memory API
- **Cognee** — in-estate only
- **Supermemory** — proprietary memory + knowledge API
- **Letta** — agent harness
- **LangMem** — no release since Oct 2025

L6 Retrieval & knowledge stores

derived, entitlement-filtered, rebuildable index

Hybrid search inside a platform you already run; a dedicated vector engine only when a load test proves the need.

- **Elasticsearch** — hybrid + document-level security
- **Milvus / Zilliz** — very large corpora
- **PostgreSQL + pgvector** — where Postgres is standard
- **Qdrant** — dedicated engine after a load test
- **MongoDB Vector Search** — GA self-managed too
- **Pinecone** — managed · BYOC
- **Weaviate** — licence in transition
- **S3 Vectors** — AWS cost tier · no BM25
- **turbopuffer** — many tenants · BYOC only
- **Chroma** — prototypes and harnesses

L7 Retrieval optimisation

pinned embed + rerank, hybrid fusion, eval gate

Choose models by in-domain evaluation; pin every version; keep raw text so a model switch is a re-embed, not a rebuild.

- **Sentence Transformers** — self-hosting and fine-tuning toolkit
- Google Cloud — **Gemini Embedding 2** — EU endpoint excludes the UK
- **Cohere Embed 5 + Rerank** — private deployment
- **OpenAI text-embedding-3** — text baseline
- **Jina AI** — Elastic-owned · CC-BY-NC weights
- **Qwen3 Embedding** — self-host after review
- **Voyage AI** — MongoDB-owned
- **NVIDIA NeMo Retriever** — NVIDIA estates

L8 Ingestion & data preparation

approved sources, ACLs and lineage on every chunk

A built control envelope (source register, classification, parse manifest, lineage, incremental indexing) around replaceable parsers.

- **Docling** — default engine · LF AI & Data
- **Unstructured** — ACL-aware connectors
- Google Cloud — **Google Document AI** — processor region confirmed
- **Mistral OCR 4.1** — pin the model ID
- **LlamaParse** — parse and extract only
- **Firecrawl** — AGPL server · cloud with ZDR
- **Reducto** — hard documents
- **Apify** — public data only
- **Crawl4AI** — pre-1.0 · pilots
- **MinerU** — licence thresholds apply

EVALUATION & OBSERVABILITY PLANE — L9 joins C8 as one evidence plane with two owners

L9 Evaluation & observability plane

evidence for every layer, from day one

One firm-owned OpenTelemetry spine, Git-versioned evaluation sets, one platform of record, and two red-team tools, one independent of the model vendor.

- **MLflow GenAI** — where an ML platform exists
- **Langfuse** — ClickHouse-owned · self-host
- **LangSmith** — LangGraph estates · BYOC, EU
- **Opik** — Comet · Apache-2.0
- **Arize AX** — Dynatrace-owned
- **DeepEval** — CI metric library
- **Promptfoo** — red-teaming · OpenAI deal announced
- **Arize Phoenix** — Dynatrace-owned · ELv2
- **W&B Weave** — CoreWeave-owned
- **Braintrust** — eval-led teams
- **Datadog Agent Observability** — Datadog APM estates

The architecture in one sentence

Build a firm-owned control and evidence plane first; run regulated work as deterministic workflows with one bounded model step and a named human approver; consume models as a two-vendor portfolio through the primary cloud; treat every product beneath that plane as replaceable.

- **One gateway of record** for all model, tool and agent traffic: in region, deployed twice, failing closed.
- **Every agent a registered identity**, acting for a named person through short-lived tokens; deny by default.
- **One privacy service** at six points: ingestion, prompt, tool results, output, memory writes and trace export.
- **Git as the configuration of record**: prompts, model pins and tool lists released as one approved manifest.
- **Build order:** governance and evaluation, then models through the gateway, retrieval, workflows, tools — memory last.

How to read it

- Tiers are for a regulated asset manager, scored on one eight-criterion rubric with regulated-FS weights. A Strategic tier carries its condition; the condition is the decision.
- Cloud-tagged items are alternatives chosen by primary cloud, not a shopping list.
- Disclosure: researched and drafted with an Anthropic model. Anthropic items were scored on the same rubric; their tiers were set by the reader, and an independent alternative is named for each.
- Personal research, not any firm's platform. Every tile is backed by sourced facts in the dataset and the product appendix.

Source: Enterprise GenAI Full-Stack Architecture review, October 2026 — 05_Data/products.json (tiers), 06_References/bibliography.xlsx (sources). Product names are trademarks of their owners; no logos used.